1. ConvertAudioEngine-DDP51.ps1 v3.0.5

> Converts all audio tracks (7.1 → downmix, 5.1 → re-encode, passthrough where applicable).
> Only stereo and commentary tracks are removed.

- Downmixes **7.1 audio** to **DDP 5.1 (1152k)** via ITU-R BS.775 pan matrix
- Re-encodes **5.1 audio** to **DDP 5.1 (768k)**; DTS-HD MA/HRA re-encodes at **1024k**
- Passthrough for **EAC3 Atmos**, **EAC3 5.1**, and **TrueHD 5.1**
- All **2.0 stereo tracks are removed** — stereo output is handled by ConvertAudioEngine-Stereo
- Maintains perfect sync on long-duration files
- Normalizes malformed channel layouts
- Removes 2-channel commentary tracks

Requirements

PowerShell 7.6 - [Click here to Download PS 7.6 (Windows x64 .msi)]
FFmpeg 8.1 - https://ffmpeg.org/download.html

Quick Start (Windows 11)

1. Place `ConvertAudioEngine-DDP51.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

> pwsh -ExecutionPolicy Bypass -File .\ConvertAudioEngine-DDP51.ps1 ".\YourMovie.mkv"

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, `ffmpeg`, and `ffprobe` in the same folder.
2. Make the binaries executable (first time only):

> chmod +x ffmpeg ffprobe

3. Run:

> pwsh -File ./ConvertAudioEngine-DDP51.ps1 "./YourMovie.mkv"

1 To Adjust Bitrate (line 66-70)

```
You can safely adjust these numbers, within **'....k'**
Common EAC3 5.1 bitrates: 384, 448, 512, 640, 768
Higher pipeline bitrates: 1024, 1152, 1280, 1408, 1536

$BitRateConfig = @{
    Downmix51   = '1152k'   # << Change your Bitrate
    ReEncode51  = '768k'    # << Change your Bitrate
    EncodeDTSHD = '1024k'   # << Change your Bitrate
    # PassCopy51 entries added in Keep71 script
```

Remember to save the file after making changes.
All three values can be adjusted safely — changes apply automatically throughout the script.

Key	Default	Used for
Downmix51	1152k	All 7.1 sources downmixed to 5.1
ReEncode51	768k	5.1 sources re-encoded to DD+
EncodeDTSHD	1024k	DTS-HD MA and HRA sources (5.1 and 7.1)

🎧 Supported Audio Codecs

The engine supports and processes the following codec families:

Codec Family	Formats Included
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm_s16le, pcm_s24le, pcm_f32le, pcm_f32be
FLAC	All FLAC channel layouts

2. FFmpeg Global Options

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- `-y` : Overwrites the output file without asking.
- `-loglevel error` : Suppresses all FFmpeg output except actual errors.
- `-loglevel warning` : Shows warnings and errors, but hides info, debug, and trace output.
- `-analyzeduration 200M` & `-probesize 200M`
  - **What:** Increases the initial data analysis buffer to 200 Megabytes.
  - **Why:** High-bitrate audio like TrueHD with Atmos metadata or AAC 7.1 requires a deep probe to fully parse the channel layout. Standard 5MB, 10MB, 20MB probes often misidentify these tracks, causing the encode to fail or use the wrong channel map.
- `-err_detect ignore_err`
  - **What:** Instructs FFmpeg to bypass minor bitstream errors.
  - **Why:** Prevents a single corrupted audio frame (common in digital rips) from aborting a multi-hour conversion process.
- `-drc_scale 0`
  - **What:** Disables Dynamic Range Compression (DRC).
  - **Why:** Ensures the output audio maintains the full dynamic range of the source. It prevents the encoder from "flattening" the sound or lowering the volume of loud peaks.

2.5 ScanThrottle + ThreadCount

Sets how many CPU threads the encoder uses (line 47).

```
$ThreadCount = 8    # User-adjustable (4-16).
```

Limits how many peak-scan tasks run at the same time (line 49).

```
$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount /
$ThreadCount))
# Recommended overrides: 1, 2, 3, or 4
```

What ScanThrottle does:

- Limits parallel SPN scans to avoid CPU overload.
- Keeps the system responsive during peak detection.
- Does not affect encoding speed.
- SSD/NVMe: safe to override up to 4. Spinning disk: keep at 1.

3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero`
  - **What:** Forces all stream timestamps to start at absolute zero.
  - **Why:** Many digital files contain negative initial "Presentation Time Stamps" (PTS); this clamps them to zero to prevent audio lag or lead.

- `-max_muxing_queue_size 14000` *(1024-100000) recommended range
  - **What:** Increases the muxing buffer headroom.
  - **Why:** Audio down-mixing and re-encoding (AAC 7.1, TrueHD 7.1, ...) can run slower than the muxer, causing FFmpeg's queue to overflow and trigger errors. 14000 provides enough headroom to prevent these overflows on **all movie lengths.**

- `-map 0:v? -c:v copy`
  - **What:** Maps all video tracks and sets them to stream-copy mode.
  - **Why:** Ensures zero quality loss for the video while moving at maximum speed.

- `-map_metadata 0 -map_chapters 0`
  - **What:** Copies global metadata and chapter markers from the source.
  - **Why:** Preserves the original file's organizational structure (Title, Year, Scene markers).

3.2 Safe Peak Normalizer (SPN)

- **What:** Safe Peak Normalizer checks each audio track before processing and only steps in when the volume is too close to clipping (above –0.5 dBFS).
- **Why:** Some movies arrive with peaks already above 0 dBFS. When that happens, even simple mixing of the channels can push those peaks higher and cause distortion in the final encode. SPN catches this early so the issue never reaches the rest of the audio processing.
- **How:** Safe Peak Normalizer runs right after the script decides each track's action (Encode, Downmix, etc.). If the peak scan shows the track is too hot, a light loudness-correction step is added at the very start of that track's filter chain.
- **Clean Sources:** Nothing is changed — the original loudness and dynamics stay exactly the same.
- **Passthrough tracks:** — Peak detection is skipped entirely. Copied streams cannot have filters applied.
- **Dirty Sources:** SPN fixes unsafe peaks without making the movie sound compressed or "over-normalized."
- **Result:** More consistent, safer audio across all movies, keeping clean sources untouched.

Threshold (line 53)

These are the values Safe Peak Normalizer uses to decide when to step in.

```
$PeakThresholdDB = -0.5 # SPN: loudnorm triggers when source peak exceeds this
(dBFS)
```

- Valid range: **−1.5 dBFS to −0.3 dBFS** (recommended: −0.5 dBFS)
- Lower values (e.g. −1.5) normalize more tracks — more conservative.
- Higher values (e.g. −0.3) normalize fewer tracks — only the hottest sources.

Setting	Value
loudnorm target	−23 LUFS
Loudness range target	LRA=11 LU
True peak ceiling	−1.5 dBTP

Three-Stage SPN Processing

The audio engine processes each audio track in three phases, with SPN peak scans running in parallel during Phase B to keep multi-track processing fast.

```
3-phase SPN Architecture (Phase A > Phase B > Phase C)
Input MKV >> Probe
     > Phase A  (rule matching + malformed-layout guards + commentary removal)
     > Phase B  (parallel SPN peak scans)
     > Phase C  (sequential merge + NeedsNormalization resolution)
     > FFmpeg command builder
       >> Output MKV
```

Phase A, Sequential rule matching
Every audio stream is probed and matched against the rule table. Malformed-layout guards and commentary detection run here. 2.0 tracks are marked Removed. Surviving tracks receive a pending action (Passthrough, Downmix, Encode).

Phase B, Parallel SPN peak scans
All non-Removed, non-Passthrough tracks are scanned in parallel for peak levels using **volumedetect**.

Phase C, Sequential merge
Peak results are merged back into the track list. **NeedsNormalization** is resolved from the scan data. The final track list is handed to the FFmpeg command builder.

4. Stereo (2.0) Tracks — Removed

All 2.0 stereo tracks are removed by this script, regardless of codec. Stereo output is delegated to **ConvertAudioEngine-Stereo**, which handles 2.0 encoding with the correct stereo-specific flags and bitrates. This keeps each script focused on a single job.

5. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- **`-filter:a:$i $panFilter`**
  - **What:** Applies the custom channel-mapping matrix (the Pan Filter).
  - **Why:** Necessary for downmixing 7.1 to 5.1. It ensures that side channels (SL/SR) are folded into the rear speakers rather than simply deleted.
- **`-c:a:$i eac3`**
  - **What:** Sets the codec for audio stream `$i` to Dolby Digital Plus (EAC3).
  - **Why:** Chosen for its high efficiency and broad compatibility with modern smart TVs and home theater receivers.
- **`-ac 6`**
  - **What:** Forces the output to exactly 6 channels (5.1).
  - **Why:** Applied on the 5.1 re-encode path to guarantee the output channel count is correct, even if the source reports a non-standard layout.
- **`-b:a:$i $t.Bitrate`**
  - **What:** Sets the target bitrate (e.g., 1152k for 7.1 downmix, 768k for 5.1 re-encode, 1024k for DTS-HD).
  - **Why:** High bitrates ensure the "transparency" of the encode, making it difficult to distinguish from the lossless original.
- **`-dialnorm -31`**
  - **What:** Sets Dialogue Normalization to the reference "off" position.
  - **Why:** Prevents the playback device from applying its own internal volume leveling, preserving the original mix's intended loudness.
- **`-cutoff 20000`**
  - **What:** Sets the low-pass filter frequency to 20 kHz.
  - **Why:** Preserves the high-frequency clarity of the audio that standard encoders sometimes discard to save space.
  - **Applies to:** Downmix path and 5.1 re-encode path only.
- **`-metadata:s:a:$i "title=..."`**
  - **What:** Injects a custom string into the stream title metadata.

- **Why:** Allows users to see exactly what the track is (e.g., "DD+ 5.1 Downmix (1152k)") in their media player's audio selection menu.

6. Audio Rules & Priority

The audio-engine uses a **Priority Mapping (0-100)** to evaluate tracks based on quality and codec type.

EAC3 / Atmos Rules

Priority	Condition	Action	Output
100	**Atmos (JOC)**	Passthrough	Original EAC3 Atmos — preserves object metadata
99	**EAC3 7.1**	Downmix	DDP 5.1 @ 1152k — maps 8 channels to 6 using pan filter
98	**EAC3 5.1**	Passthrough	Original EAC3 5.1
97	**EAC3 2.0**	Removed	Delegated to ConvertAudioEngine-Stereo

TrueHD Rules

Priority	Channels	Action	Output
81	5.1	Passthrough	Original TrueHD 5.1
80	7.1	Downmix	DDP 5.1 @ 1152k

AAC Rules

Priority	Channels	Action	Output
92	7.1	Downmix	DDP 5.1 @ 1152k
91	5.1	Encode	DDP 5.1 @ 768k
90	2.0	Removed	Delegated to ConvertAudioEngine-Stereo

DTS Rules

Priority	Variant	Action	Output
73	DTS-HD MA/HRA (any ch > 2)	Encode	DDP 5.1 @ 1024k
72	DTS Core (any ch > 2)	Encode	DDP 5.1 @ 768k
71	DTS-HD 2.0	Removed	Delegated to ConvertAudioEngine-Stereo
70	DTS 2.0	Removed	Delegated to ConvertAudioEngine-Stereo

PCM / FLAC Rules

Priority	Channels	Action	Output
62	7.1	Downmix	DDP 5.1 @ 1152k
61	5.1	Encode	DDP 5.1 @ 768k
60	2.0	Removed	Delegated to ConvertAudioEngine-Stereo

7. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

The Formula

> aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707*SL|BR=BR+0.707*SR
> alimiter=limit=0.948:attack=5:release=50:level=disabled:latency=1

Layout Locking (aformat)

What: Forces the decoder to output a strict 8-channel 7.1 layout *before* the Pan Filter.
Why: Some codecs (TrueHD, AAC, EAC3) may decode into non-standard 7.1 variants.
Locking the layout ensures:

- SL really is the Side Left channel
- BL really is the Back Left channel
- No channel swapping
- No silent corruption of the downmix

The Math (0.707)

What: Applies a –3 dB coefficient to the Side Surrounds (SL/SR).
Why: Prevents the side channels from overpowering the rear soundstage and avoids digital clipping.

Limiter (0.948) — For All 7.1 Codecs

alimiter=limit=0.948:attack=5:release=50:level=disabled:latency=1
Why: All 7.1 codecs can produce true peak overshoots after decoding:

- AAC 7.1 > AAC can create peaks slightly above 0 dB after decoding.
- TrueHD 7.1 > Combining similar channels can briefly push the volume higher.
- DTS-HD MA > Decoder can rebuild peaks that go a bit above the original level.
- PCM/FLAC > Short volume spikes can appear when channels are mixed together.
- EAC3 7.1 > Channels are merged into fewer speakers during downmixing.

Limiter ceiling of 0.948 provides about 0.5 dB of true-peak headroom, preventing:
1.) +0.0x dB overshoots
2.) clipping

latency=1 allows the limiter a 1-sample look-ahead window to catch post-downmix peaks without adding audible delay.

Important:

- If the source audio already contains true-peak or inter-sample peaks ("clips"), these will also appear in **analysis** of the downmix.
- The limiter prevents new clipping during downmixing and encoding, but **it does not remove overshoots inherent in the source mix**.

8. Commentary Filtering

Only 2-channel commentary tracks are removed to save space while preserving primary content.

- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** commentary, director, producer, writer, cast, behind, bonus, alt, interview.
- **Why 5.1 Commentary is NOT removed:**
  - It is extremely rare in retail media.
  - It is often hard to tell apart from legitimate "Alternate Mixes" or "Home Theater Optimized" tracks.
  - Removing it carries a high risk of "False Positives" (deleting real movie content).

9. Malformed Layout Guards

The script corrects illegal 7-channel layouts before rule matching runs:

- **AAC 7ch** → 6ch (treated as 5.1)
- **EAC3 7ch** → 6ch (treated as 5.1)
- **TrueHD 7ch** → 8ch (treated as 7.1)
- **PCM/FLAC 7ch** → 8ch (treated as 7.1)
- **AC3 7ch** → Removed (no valid AC3 7-channel layout exists)

⛔ Fallback Rules

If no specific rule matches, the following logic is applied:

Channels	Fallback Action	Bitrate
2 channels or fewer	Removed	(delegated to ConvertAudioEngine-Stereo)
3 to 6 channels	Encode	768k
more than 6 channels (7ch+)	Downmix	1152k

10. Summary Report

At === AUDIO PROCESSING SUMMARY ===, the script generates a detailed table with the following columns:

Column	Description
Index	The original stream number (0, 1, 2, 3,).
Codec	The source format (e.g., TrueHD, DTS).
Channels	The detected number of channels (e.g., 7.1, 5.1).
Action	**Green**: Passthrough, **Yellow**: Downmix, **Blue**: Encode, **Red**: Removed.
Output	The final codec and bitrate configuration.
Priority	The assigned priority level (0–100). Removed tracks show `–`.
Rule	the logical rule that was matched during processing, such as "TrueHD 7.1 Rule"

11. Re-Enabling 2.0 Encoding (Optional)

All 2.0 rules are present in the engine but set to `Action = "Removed"` by default. This is intentional — stereo output is delegated to **ConvertAudioEngine-Stereo**. If you want this script to encode 2.0 tracks instead of removing them, you can re-enable any rule by changing three fields.

Example — re-enabling AAC 2.0 at 256k:

```
# Find this block in $Rules_AAC (Rule3):
Action          = "Removed"
Bitrate         = $null
Rule            = "AAC_2.0_Removed"

# Change it to:
Action          = "Encode"
Bitrate         = "256k"
Rule            = "AAC_2.0_Encode_256k"
```

Only those three fields need to change. Nothing else in the engine requires modification.

Apply the same pattern to re-enable any other 2.0 rule:

Rule block	Location	Default	Suggested bitrate
AAC 2.0	Rule3 in `$Rules_AAC`	Removed	256k
EAC3 2.0	Rule4 in `$Rules_EAC3_Atmos`	Removed	384k
DTS-HD 2.0	Rule3 in `$Rules_DTS`	Removed	384k
DTS 2.0	Rule4 in `$Rules_DTS`	Removed	256k
PCM/FLAC 2.0	Rule3 in `$Rules_PCMFLAC`	Removed	384k

12. Non-Critical FFmpeg Warnings (Safe to Ignore)

During processing, FFmpeg may display one or more of the following messages.
These warnings are **normal**, and do not affect audio quality, sync, or re-encode.
They come from FFmpeg’s internal decoders and subtitle parsers — not from the script!

“quant_step_size larger than huff_lsbs”

- **Cause:** Printed by the TrueHD decoder during decoding (not probe).
- **Impact:** Safe to ignore. Does not affect decoding, channel layout, or the Pass+Copy path.

“Codec AVOption drc_scale … has not been used for any stream”

- **Cause:** (`-drc_scale`) applies only to AC-3/EAC3 decoders.
  FFmpeg prints this when the input track is not AC-3 (e.g., TrueHD, DTS, FLAC).
- **Impact:** Safe to ignore. The script disables DRC intentionally for consistent output.

“(Subtitle: hdmv_pgs_subtitle) unspecified size”

- **Cause:** PGS (Blu-ray) subtitles do not declare width/height in the container.
  FFmpeg warns until the first subtitle packet is decoded.
- **Impact:** Safe to ignore. Subtitles are copied untouched (`-c:s copy`).

“Assuming an incorrectly encoded 7.1 channel layout...instead of 7.1(wide)”

- **Cause:** AAC has two 7.1 layouts. Most encoders use the common (non-spec) layout.
  FFmpeg prints this when the file does not use the strict 7.1(wide) AAC layout.
- **Impact:** Safe to ignore. The script already corrects and pins the layout using (`aformat=channel_layouts=7.1`)

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.